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Directed Percolation in Temporal Networks

Physics of Complex Systems

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Based on A. Badie-Modiri et al., *Phys. Rev. Research* **4**, L022047 and *Phys. Rev. E* **105**, 054313 (2022)



Question and strategy

When can local, time-ordered interactions produce macroscopic reachability?

Can this transition show the critical scaling of **directed percolation**?

PRR

Event-graph framework

Mean-field theory

Application to real data

PRE

Numerical tests

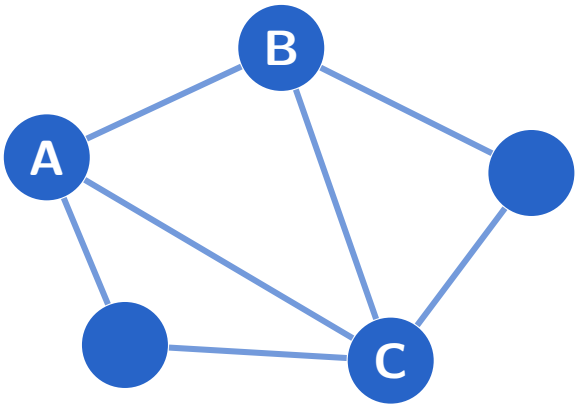
Structural and temporal heterogeneity

Critical scaling

Different thresholds — same critical scaling?

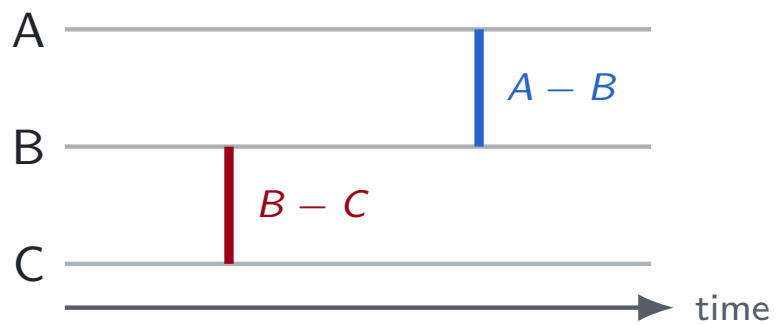
Connectivity becomes causal when links change in time

Static network



A path depends only on adjacency.

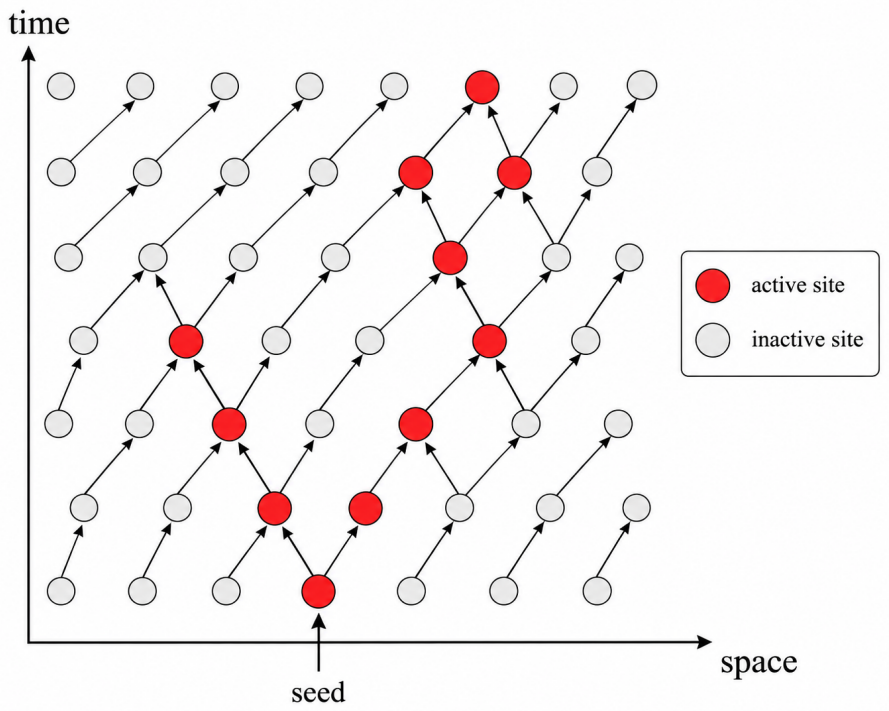
Temporal network



A valid path must also respect event order.

Aggregating over time can create paths that never existed dynamically.

Directed percolation: the reference transition



Active phase

non-zero probability of indefinite survival in the thermodynamic limit

Critical point

scale-free clusters and diverging correlations

Absorbing phase

activity dies out and cannot restart spontaneously

Criticality is encoded by scaling laws

Homogeneous initial condition

$$\rho_{\text{stat}}(\tau) \sim \tau^{\beta}, \qquad \tau > 0,$$
$$\rho(t) \sim t^{-\alpha}, \qquad \tau = 0.$$

ρ measures the fraction of active sites.

Correlation scales

$$\xi_{\parallel} \sim |\tau|^{-\nu_{\parallel}},$$
$$\xi_{\perp} \sim |\tau|^{-\nu_{\perp}}.$$

Single-seed initial condition

$$P(t) \sim t^{-\delta}, \qquad \tau = 0,$$
$$P_{\infty}(\tau) \sim \tau^{\beta'}, \qquad \tau > 0,$$
$$M(\tau) \sim (-\tau)^{-\gamma}, \qquad \tau < 0.$$

M counts all activity generated before extinction.

Different microscopic models can share these exponents: this defines a **universality class**.

Temporal paths preserve the arrow of time

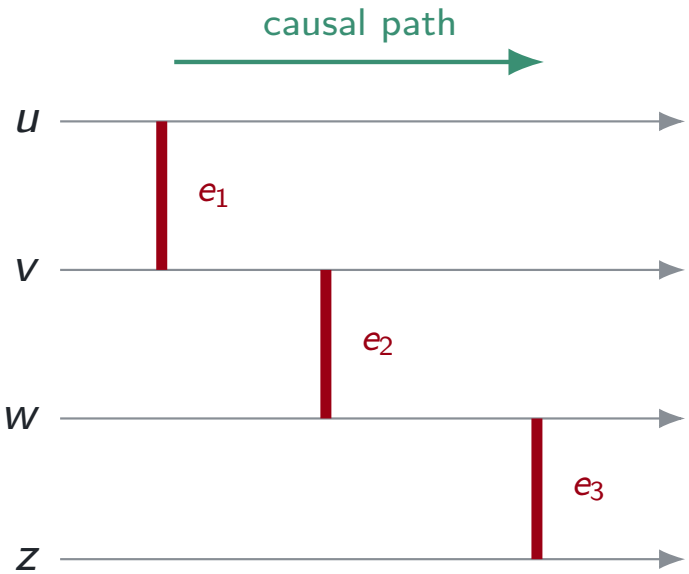
Temporal event

$$e = (u, v, t_{\text{start}}, t_{\text{end}})$$

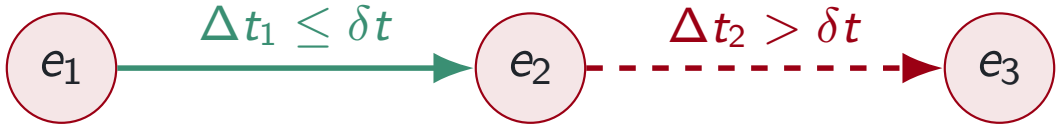
Two distinct events e and e' are adjacent, $e \to e'$, if

$$t'_{\text{start}} > t_{\text{end}},$$
$$\{u, v\} \cap \{u', v'\} \neq \varnothing.$$

A **time-respecting path** is a sequence of such adjacent events.



A waiting-time constraint controls reachability



$$e \xrightarrow{\delta t} e' \quad \Longleftrightarrow \quad e \to e' \text{ and } \Delta t(e, e') \leq \delta t.$$

Small δt

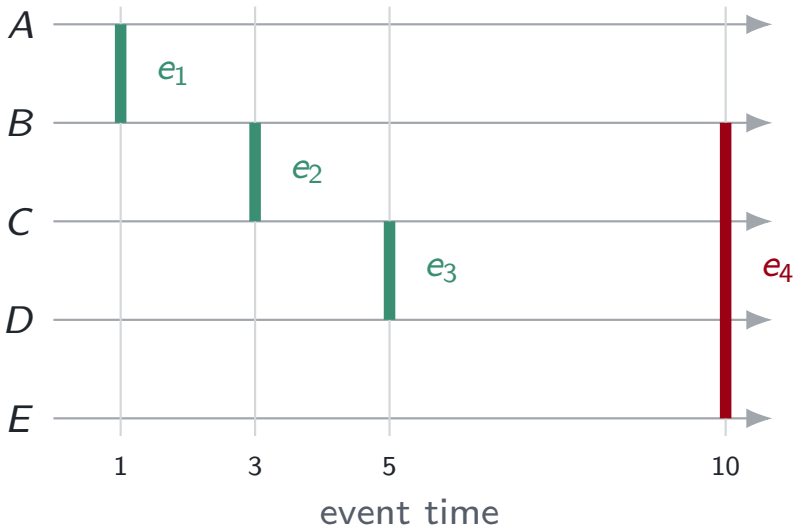
most causal chains terminate; reachability remains local

Large δt

events connect into long-lived, system-spanning paths

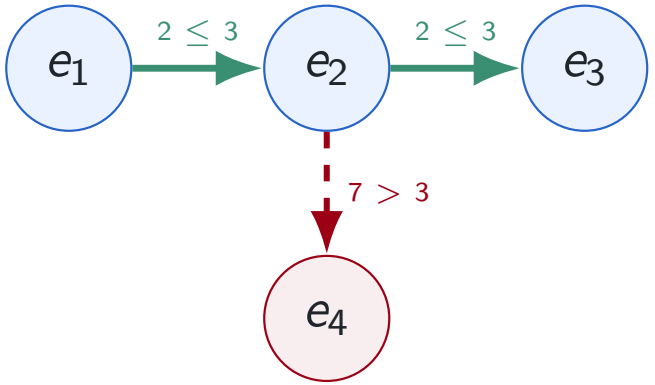
A minimal example of temporal reachability

Four instantaneous events, $\delta t = 3$



$e_1 = (A, B, 1), \quad e_2 = (B, C, 3),$
$e_3 = (C, D, 5), \quad e_4 = (B, E, 10).$

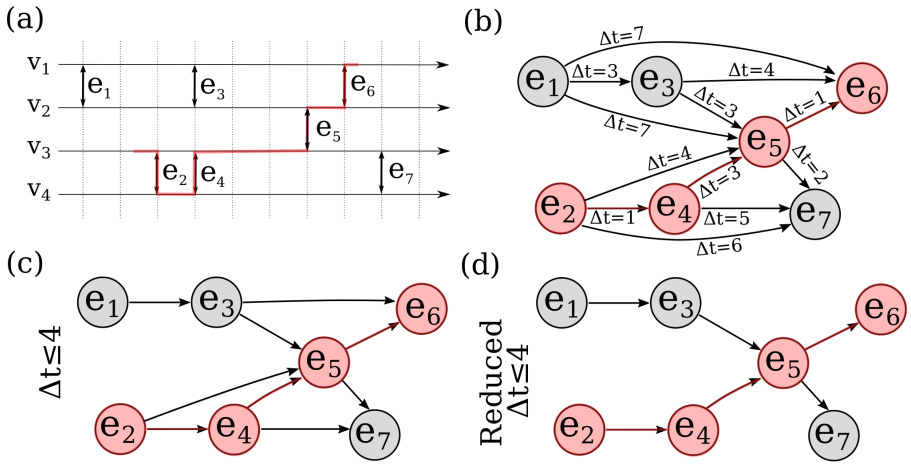
Constrained event graph



$e_1 \longrightarrow e_2 \longrightarrow e_3$

e_4 shares node B, but it occurs too late to continue the chain.

Event graphs turn temporal paths into directed paths



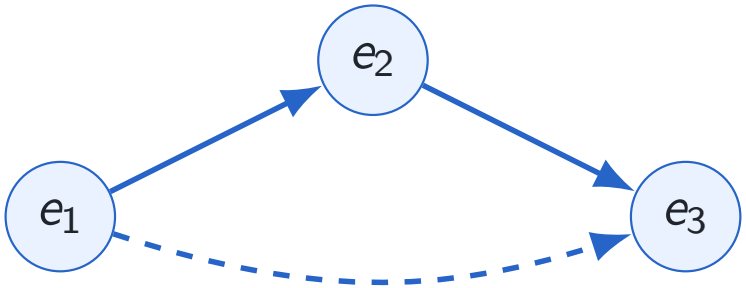
$$D = (E, E_D, \Delta t),$$
$$D_{\delta t} = (E, E_{\delta t}),$$
$$E_{\delta t} = \{(e, e') \in E_D : \Delta t(e, e') \leq \delta t\}.$$

- original events become vertices;
- temporal adjacency becomes a directed edge;
- the edge weight is the waiting time;
- paths in $D_{\delta t}$ are exactly the allowed temporal paths.

Adapted from PRR, Fig. 1.

Reduction removes redundancy, not reachability

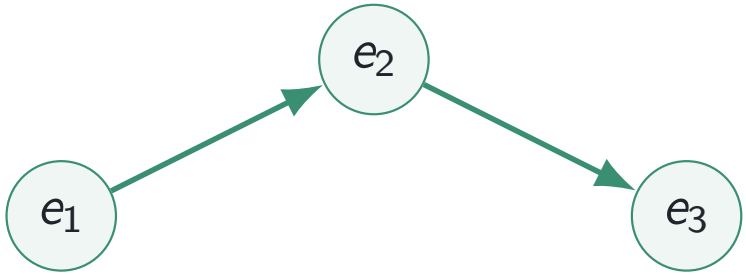
Event graph $D_{\delta t}$



Dashed edge: a redundant shortcut.

e_3 is already reachable through e_2.

Reduced graph $\widehat{D}_{\delta t}$



Reachability is unchanged.

For pairwise events, neglecting simultaneous adjacent events: $k_{\text{out}} \leq 2$.

$k_{\text{out}} = 0$	$k_{\text{out}} = 1$	$k_{\text{out}} = 2$
termination	**continuation**	**branching**

From waiting time to effective connectivity

Physical control parameter

δt

maximum waiting time

system-dependent time units

Effective local connectivity

$\widehat{q}_{\text{out}}(\delta t)$

mean excess out-degree of $\widehat{D}_{\delta t}$

dimensionless

$\delta t \longrightarrow$ allowed temporal connections $\longrightarrow \widehat{q}_{\text{out}}$

$\widehat{q}_{\text{out}}$ = average number of future continuations after following a directed edge

Distance from criticality

$$\tau = \widehat{q}_{\text{out}} - \widehat{q}^{c}_{\text{out}}$$

Mean-field theory later gives $\widehat{q}^{c}_{\text{out}} = 1$.

Observables of a reachable cluster

Starting from a random source event e:

M — mass

number of events in the causal out-component

V — volume

number of distinct physical nodes reached

T — lifetime

temporal span

$t_{\max}(E_e^{\rightarrow}) - t_{\min}(E_e^{\rightarrow})$

$$P(t) = \Pr(\text{a path from } t_0 \text{ survives beyond } t_0 + t).$$

M, V and T are three projections of the same causal out-component; $P_\infty = \lim_{t\to\infty} P(t)$ is an alternative order parameter.

Homogeneous occupation defines a density order parameter

Start from all events before t_0 occupied.

$\rho(t)$ is the occupation density measured in a temporal band around t.

$$\rho_{\mathrm{stat}}(\tau) = \lim_{t\to\infty} \rho(t).$$

$\rho_{\mathrm{stat}} = 0$ in the absorbing phase;
$\rho_{\mathrm{stat}} > 0$ in the active phase.

An external field h spontaneously activates events and removes the strictly absorbing state.

$$\chi(\tau, h) = \frac{\partial \rho_{\mathrm{stat}}}{\partial h}.$$

The peak or divergence of χ helps locate the transition.

From microscopic temporal rules to branching statistics

Static topology

q_l, q_r: endpoint excess degrees

candidate future channels

+

Temporal dynamics

inter-event and residual waiting times; δt

which channels activate in time

$k_{\text{out}} = 0$

termination

$k_{\text{out}} = 1$

continuation

$k_{\text{out}} = 2$

branching

Local branching probabilities

$$P_{\text{out}}(0 \mid q_l, q_r), \quad P_{\text{out}}(1 \mid q_l, q_r), \quad P_{\text{out}}(2 \mid q_l, q_r)$$

Static topology provides the possible channels; temporal dynamics determines which become causal continuations.

After averaging over endpoint degrees to form $P(k_{\text{in}}, k_{\text{out}})$, and then accounting for arrival bias, these give Q_0, Q_1, Q_2 for mean field.

The mean-field approximation



Exact representation: time-respecting paths in the temporal network are directed paths in the constrained event graph, and the reduction preserves reachability.

Mean-field closure: replace the reduced event graph by a locally tree-like random digraph with the same joint distribution $P(k_{\mathrm{in}}, k_{\mathrm{out}})$.

Retained

- in/out-degree statistics;
- average branching;
- dependence on δt through those statistics.

Neglected

- higher-order correlations;
- causal motifs;
- spatial organization beyond degree statistics;
- branch-meeting correlations.

Three branching probabilities determine local growth

After following a directed edge, define

$$Q_i \equiv Q_{\text{out}}(i) = \Pr(\text{excess out-degree} = i), \qquad Q_0 + Q_1 + Q_2 = 1.$$

probability	change in active branches	meaning
Q_0	-1	termination
Q_1	0	continuation
Q_2	$+1$	branching

Under homogeneous mixing, overlap between branches produces the quadratic correction:

$$\partial_t \rho = (Q_2 - Q_0)\rho - (Q_1 + 2Q_2)\rho^2.$$

The rate equation gives the mean-field threshold

The mean excess out-degree is

$$\widehat{q}_{\text{out}} = Q_1 + 2Q_2.$$

Using $Q_0 + Q_1 + Q_2 = 1$,

$$Q_2 - Q_0 = Q_2 - (1 - Q_1 - Q_2) = \widehat{q}_{\text{out}} - 1.$$

Therefore

$$\boxed{\partial_t \rho = \tau\rho - \widehat{q}_{\text{out}}\rho^2}, \qquad \tau = \widehat{q}_{\text{out}} - 1.$$

Mean-field critical point: $\tau = 0 \Longrightarrow \boxed{\widehat{q}^c_{\text{out}} = 1}$.

$\tau < 0$	$\tau = 0$	$\tau > 0$
absorbing state	critical point	active state

Two mean-field exponents follow immediately

At criticality: $\tau = 0$

$$\partial_t \rho = -\widehat{q}_{\text{out}} \rho^2$$

$$\rho(t) = \frac{\rho_0}{1 + \widehat{q}_{\text{out}} \rho_0 t} \underset{t \to \infty}{\sim} t^{-1}.$$

$$\boxed{\alpha = 1}$$

In the active phase: $\tau > 0$

$$0 = \tau \rho_{\text{stat}} - \widehat{q}_{\text{out}} \rho_{\text{stat}}^2$$

$$\rho_{\text{stat}} = \frac{\tau}{\widehat{q}_{\text{out}}} \sim \tau^1.$$

$$\boxed{\beta = 1}$$

These are the DP exponents above the upper critical dimension $d_c = 4$.

One example: the mean cluster mass

$$M \sim (-\tau)^{-\gamma}, \qquad \tau \to 0^-$$

M is the mean total number of reachable events generated by one source before its activity dies out.
Here M counts **events**, not distinct physical nodes.

For every fixed $\tau < 0$, the process eventually dies out and the mean cluster mass is finite. As the critical point is approached from below, however, the mean cluster mass diverges.

Branching-process intuition

$$M = 1 + z\left(1 + \widehat{q}_{\text{out}} + \widehat{q}_{\text{out}}^2 + \cdots\right), \qquad z = \langle k_{\text{out}} \rangle$$
$$= 1 + \frac{z}{1 - \widehat{q}_{\text{out}}} = 1 + \frac{z}{-\tau} \sim (-\tau)^{-1}.$$

$$\gamma = 1$$

Finite-size scaling as a numerical test

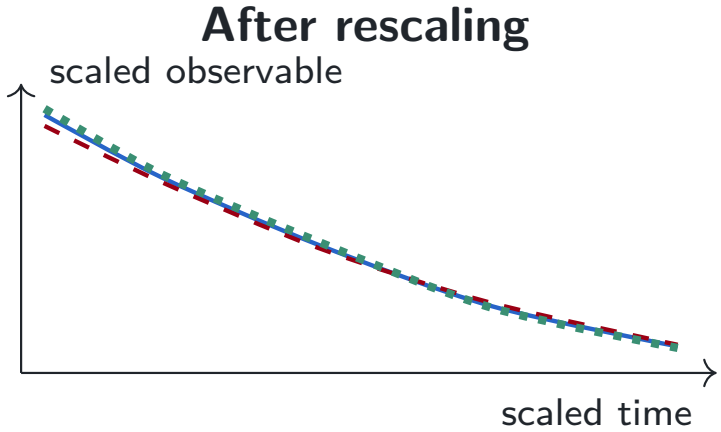
For the density at criticality,

$$\ell^{\beta/\nu_\perp}\rho(t,\ell) = F\left(t\,\ell^{-\nu_\parallel/\nu_\perp}\right).$$

Since $N \propto \ell^d$,

$$\boxed{N^{\beta/(d\nu_\perp)}\rho(t,N) = F\left(tN^{-\nu_\parallel/(d\nu_\perp)}\right).}$$

Analogous rescalings are used for $\widehat{P}(t)$, $M(t)$ and $V(t)$.



Correct δt_c + correct exponents $\Longrightarrow$ collapse across N.

A quantitative collapse across several sizes is a stronger test than observing one approximate power law.

PRE: building the synthetic models

1. Static base network

random 9-regular;
Erdős–Rényi $\langle k \rangle = 8$;
lattices $d = 1, \ldots, 4$

→

2. Temporal activation on each link

Poisson;
bursty renewal;
Hawkes

all normalized to $\langle \Delta t \rangle = 1$

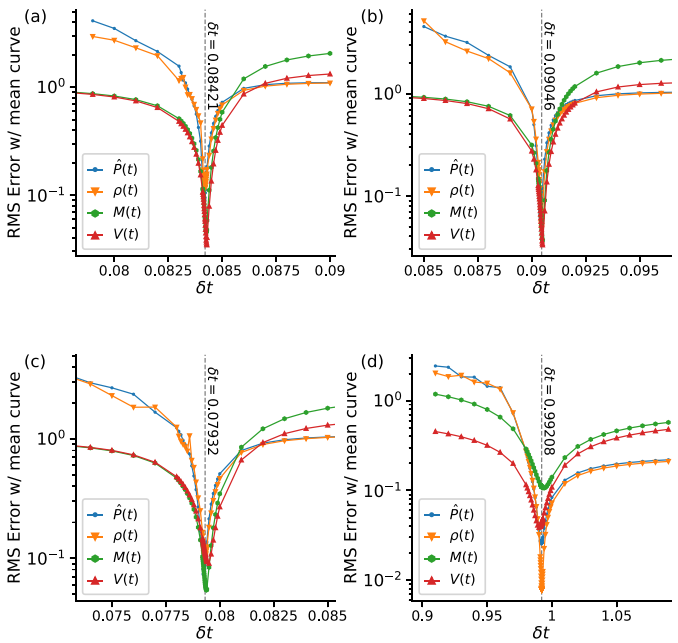
→

3. Reachability analysis

generate all time-stamped events first;
then choose δt
and construct $D_{\delta t}$

$\Delta t_i = t_{i+1} - t_i$ is an inter-event time generated by the activation process; δt is the maximum waiting-time cutoff imposed afterwards on a temporal path.

PRE: locating the critical waiting time



candidate δt

↓

rescale trajectories for every N

↓

log-RMS collapse error for each observable

↓

sum errors of $\widehat{P}$, ρ, M, V

↓

$\min E(\delta t) \implies \widehat{\delta t_c}$

Each point in Fig. 3 is a collapse error across system sizes.

Minimum total error $\implies$ estimate of δt_c.

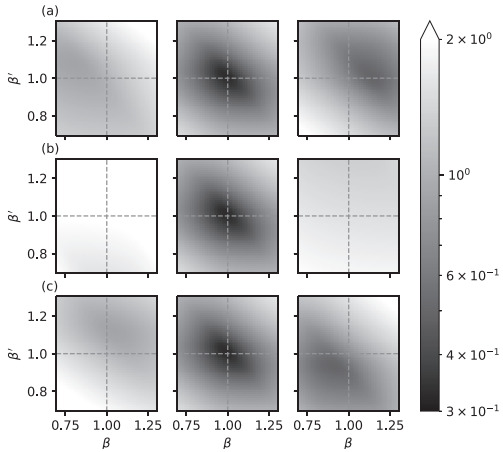
PRE, Fig. 3 and Sec. III B.

PRE: checking the critical exponents

Expected DP exponents helped estimate δt_c; the PRE then tests them independently.

CHECK 1 — Free-exponent collapse

Repeat the collapse with all five parameters free: $(\beta, \beta', \nu_{\parallel}, \nu_{\perp}, \delta t_c)$.
Does the error minimum remain close to DP?



Example: ER + Poisson. Darker = smaller collapse error. Shown are 2D slices of the five-dimensional error landscape.

Low-error region near $(\beta, \beta', \nu_{\parallel}, \nu_{\perp}) = (1, 1, 1, 1/2)$: compatible with mean-field DP.

CHECK 2 — Direct time-decay fits

At the estimated δt_c, fit the time dependence directly:

$$\rho(t) \sim t^{-\alpha}, \qquad \widehat{P}(t) \sim t^{-\delta}.$$

Direct check of the decay exponents at the estimated threshold.

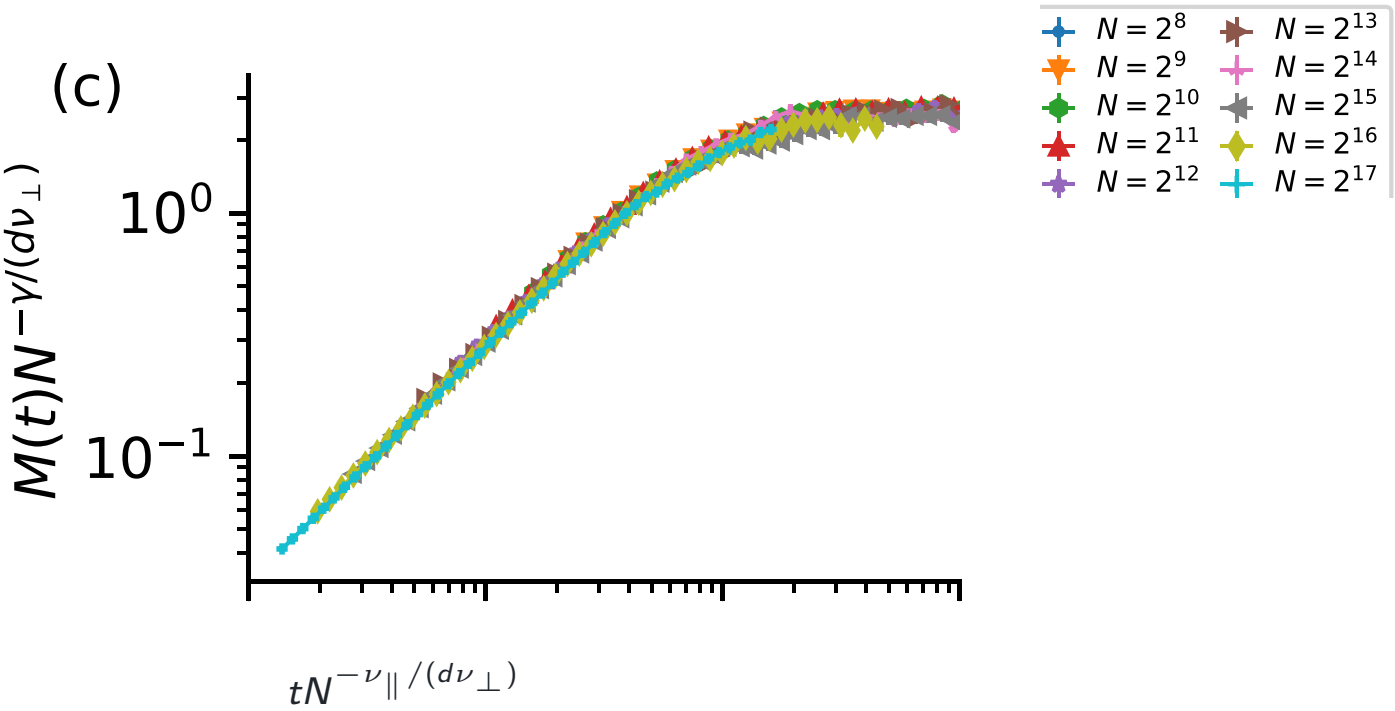
	$\widehat{\alpha}$	$\widehat{\delta}$
9-regular + Poisson	1.010	0.995
mean-field DP	1	1

Random-network cases are compared with mean-field DP; finite-dimensional lattices with the appropriate dimension-dependent DP exponents.

Across 26 configurations, thresholds shift strongly; critical scaling remains compatible with the appropriate DP regime.

PRE, Fig. 4, Fig. 6 and Table I.

A successful finite-size collapse



Example: random 9-regular + bursty renewal, $\gamma = 2.8$.

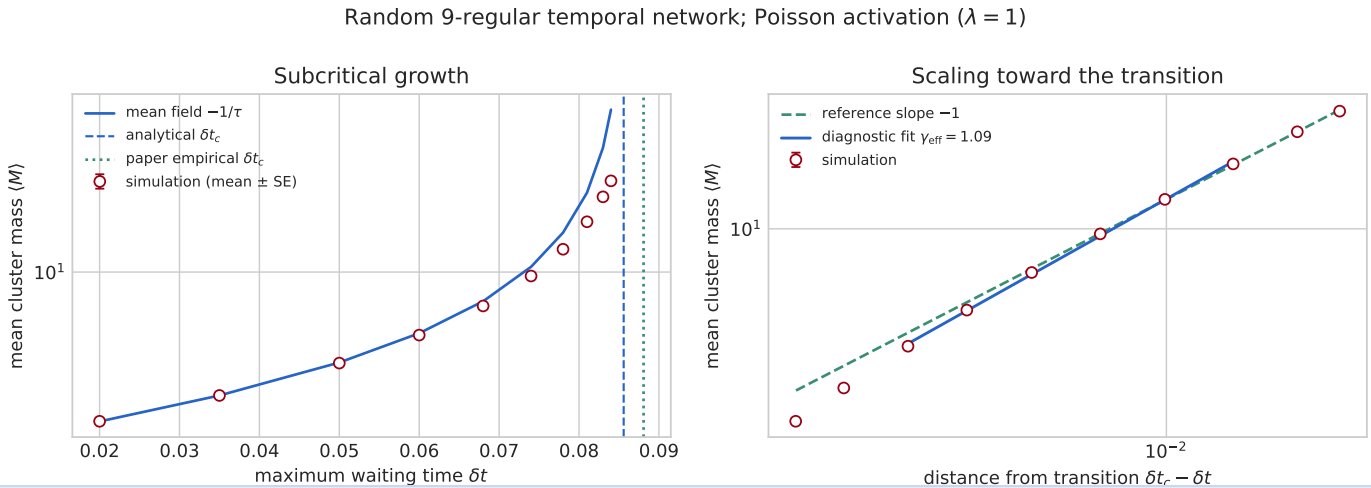
Different colours: different system sizes N.

Finite-size rescaling collapses the $M(t)$ trajectories onto one scaling function.

Correct threshold + correct exponents $\Longrightarrow$ size-independent scaling curve.

PRE, Fig. 5(c).

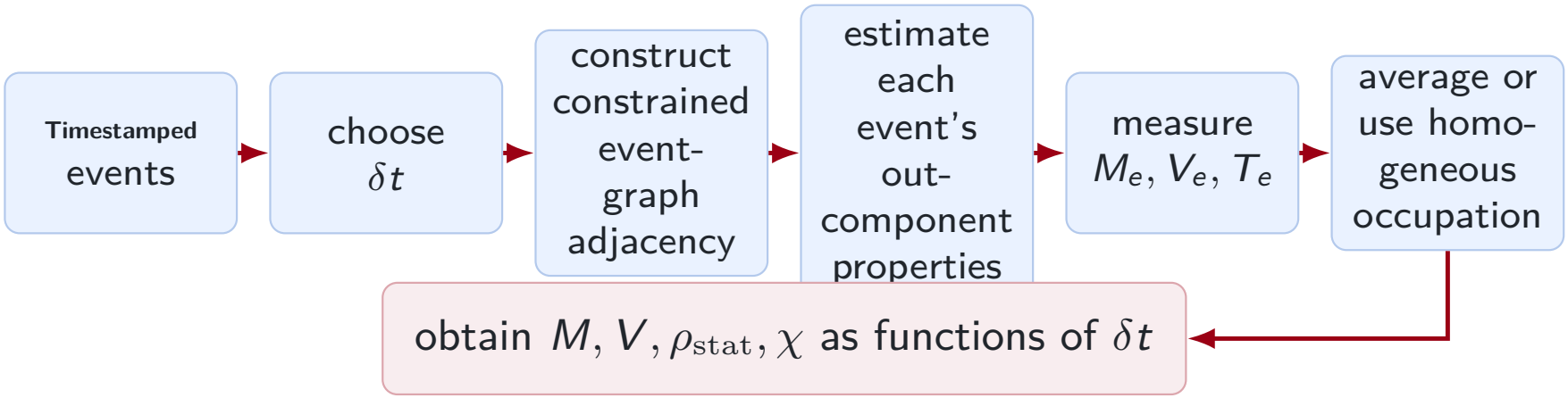
Our laptop-scale consistency check



$N = 4096$, 128 realizations, 1024 sources per realization. **Diagnostic fit:** $\gamma_{\text{eff}} \simeq 1.09$.

Subcritical expectation: $M \sim (\delta t_c - \delta t)^{-1}$. **Consistency check, not a precision measurement:** δt_c is taken from the paper and no full scaling collapse is attempted.

From empirical events to reachability



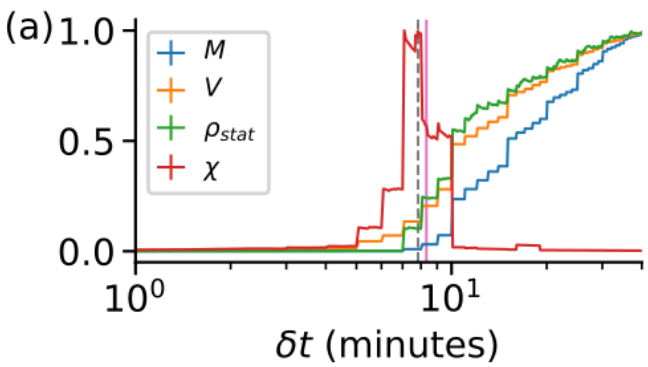
Air and Helsinki

Directed, non-instantaneous events. Waiting time runs from the end of one flight/trip to the beginning of the next.

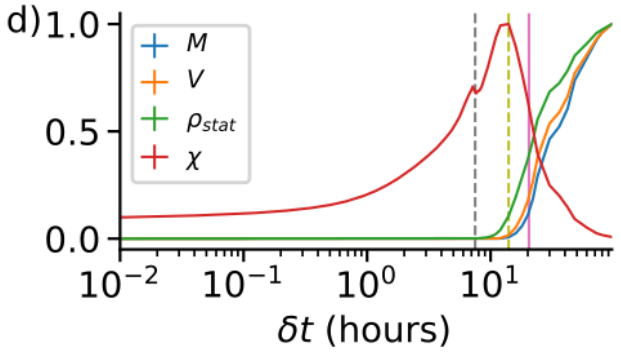
Twitter and mobile calls

Treated as undirected, instantaneous events between the two users or subscribers.

Transitions in real temporal networks



Air transport: Reachability onset near 470 s.



Mobile-phone calls: Onset near 7.5 h; susceptibility peaks near the transition.

	Air	Helsinki	Twitter	Mobile
characteristic onset	470 s	670 s	25 min	7.5 h

In every system, M, V and ρ_{stat} rise over a characteristic waiting-time range while χ peaks. Curves are rescaled to $[0, 1]$, so their vertical magnitudes are not comparable.

PRR, Fig. 3(a)–(d).

What do the results establish, and what remains open?



Synthetic models — strong evidence

- Finite-size scaling and exponent estimates are broadly compatible with the appropriate DP regime across the tested models.
- Topology and temporal dynamics can strongly shift the threshold, while the critical exponents are much more robust.

Strongest burstiness $\longrightarrow$ *larger uncertainty.*

Empirical networks — transition, but not a universality test

- Clear reachability transitions are observed in four real systems.
- The empirical analysis does not provide the same precision finite-size-scaling test of universal DP exponents.

What was not tested?

- Inter-link temporal correlations and temporal motifs were not systematically included.
- Heavy-tailed degree distributions and more complex static topologies were not fully explored.
- Different spreading mechanisms may require another definition of event adjacency, or a probabilistic rather than hard waiting-time cutoff.

Thank you for your attention!

Backup slides

Notation at a glance

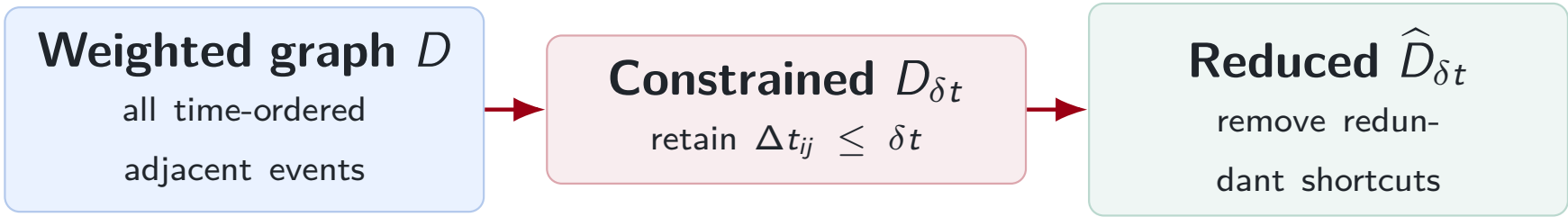
$G = (V, E, \mathcal{T})$	temporal network
$e = (u, v, t_{\text{start}}, t_{\text{end}})$	temporal event
$D = (E, E_D, \Delta t)$	weighted event graph
$D_{\delta t}$	waiting-time constrained event graph
$\widehat{D}_{\delta t}$	reduced constrained event graph
$Q_{\text{out}}(k)$	excess out-degree probability
$\tau = \widehat{q}_{\text{out}} - \widehat{q}^{,c}_{\text{out}}$	distance from criticality
$\widehat{q}^{,c}_{\text{out}} = 1$	mean-field critical value
$\rho(t), \rho_{\text{stat}}$	occupation density and stationary density
$M, V, T, P(t)$	mass, volume, lifetime and survival probability

Event graph: formal definition and exact properties

For instantaneous events $e_i = (u_i, v_i, t_i)$ and $e_j = (u_j, v_j, t_j)$,

$$e_i \longrightarrow e_j \quad \Longleftrightarrow \quad \{u_i, v_i\} \cap \{u_j, v_j\} \neq \varnothing, \qquad t_j > t_i,$$

with edge weight $\Delta t_{ij} = t_j - t_i$.



- Time strictly increases along every edge, so the event graph is a DAG.
- $D_{\delta t}$ and $\widehat{D}_{\delta t}$ have the same reachability.
- For pairwise, non-simultaneous events, $k_{\text{in}}, k_{\text{out}} \leq 2$ in the reduced graph.

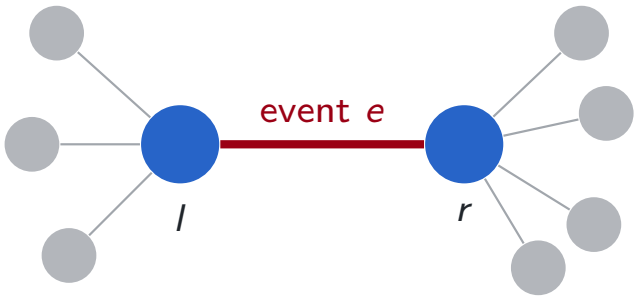
PRR, definitions and Supplemental Material.

Two different notions of degree

Base-network excess degrees

$$q_l,\ q_r$$

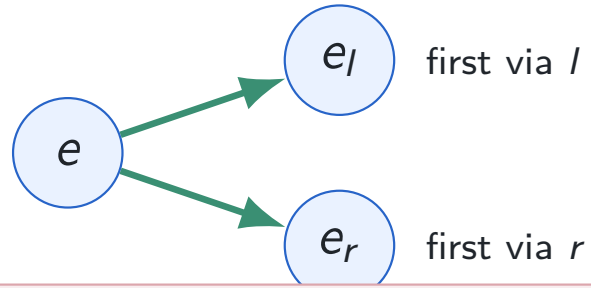
Excess degrees of the endpoints l and r in the underlying static network. They count candidate adjacent links and may exceed two.



Reduced-event-graph out-degree

$$k_{\text{out}} \in \{0, 1, 2\}$$

Actual out-degree after retaining only the first useful future event through each endpoint. Hence it is at most two when simultaneous adjacent events are negligible.



The static excess degrees count candidate physical links and may be much larger than two. The bound $k_{\text{out}} \leq 2$ arises only after temporal ordering and reduction of the event graph; it is not a bound on the underlying static-network degree.

Inter-event and residual waiting times

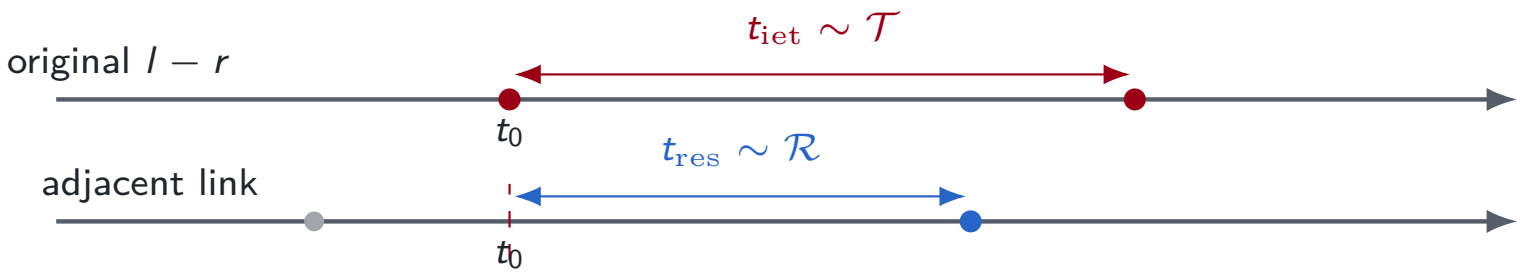
Consider the event $e = (l, r, t_0)$. At the observation time t_0, the original link and its adjacent links are in different renewal states.

Original link $l - r$
At t_0 it has just activated. Its next activation follows a full inter-event time:

$t_{\text{iet}} \sim \mathcal{T}$.

Adjacent links
At t_0 they are observed at an arbitrary point in their renewal cycle. Their next activation follows a residual time:

$t_{\text{res}} \sim \mathcal{R}$.



The original link starts a new renewal interval at t_0; adjacent links are inspected partway through an interval.

PRR Supplement, discussion before Eq. (1).

Residual waiting-time distribution

For a stationary renewal process, let $\mathcal{T}(t)$ be the inter-event-time density and $\mathcal{R}(t)$ the residual waiting-time density observed from a random time.

Explanatory renewal-theory relation

$$\mathcal{R}(t) \quad = \quad \frac{1}{\langle t_{\mathrm{iet}} \rangle} \int_t^\infty \mathcal{T}(u)\, du.$$

A random observation time is more likely to fall inside a long renewal interval. Therefore the remaining time is generally not distributed like a fresh inter-event time.

For a Poisson process, $\mathcal{T} = \mathcal{R}$ because of memorylessness; for a non-Poisson renewal process they generally differ.

PRR Supplement, discussion before Eq. (1).

Probability of no continuation

Out-degree zero means that no usable future event occurs through either endpoint of $e = (l, r, t_0)$ within the cutoff δt.

$$P_{\text{out}}(0 \mid q_l, q_r) = \Pr(t_{\text{res}} > \delta t)^{q_l+q_r} \Pr(t_{\text{iet}} > \delta t).$$

Adjacent links

$\Pr(t_{\text{res}} > \delta t)^{q_l+q_r}$

None of the $q_l + q_r$ adjacent physical links fires within δt.

Original link $l - r$

$\Pr(t_{\text{iet}} > \delta t)$

The link that generated the event at t_0 does not reactivate within δt.

Static topology fixes the number of candidate links; $\mathcal{T}$, $\mathcal{R}$ and δt determine whether they fire in time.

PRR Supplement, Eq. (1).

Probabilities for zero, one and two continuations

Two outgoing continuations occur only if at least one usable event appears through the l side **and** at least one through the r side, both before δt and before the original $l - r$ link reactivates.

$$P_{\text{out}}(2 \mid q_l, q_r) = \int_0^\infty [1 - \Pr(t_{\text{res}} > \delta t \vee t_{\text{res}} > t)^{q_l}] \times [1 - \Pr(t_{\text{res}} > \delta t \vee t_{\text{res}} > t)^{q_r}] \Pr(t \sim \mathcal{T})\, dt.$$

The remaining case is

$$P_{\text{out}}(1 \mid q_l, q_r) = 1 - P_{\text{out}}(0 \mid q_l, q_r) - P_{\text{out}}(2 \mid q_l, q_r).$$

0 — no useful continuation

1 — one effective branch

2 — one continuation through each endpoint

The probabilities depend jointly on q_l, q_r and on $\mathcal{T}, \mathcal{R}, \delta t$.

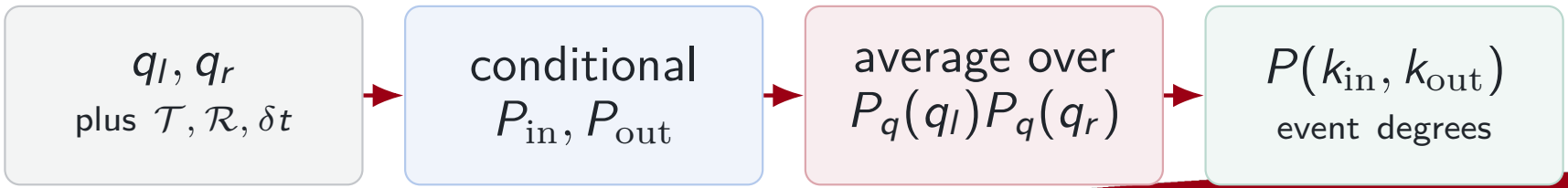
PRR Supplement, Eqs. (2)–(3).

Averaging over endpoint degrees

Average the conditional event-graph probabilities over the base-network excess-degree distribution P_q:

$$P(i,o) = \sum_{q_l,q_r=1}^{\infty} P_{\text{in}}(i \mid q_l, q_r) P_{\text{out}}(o \mid q_l, q_r) P_q(q_l) P_q(q_r).$$

Here we average over the possible static-network excess degrees of the two endpoints. The result $P(i,o)$ is the joint in/out-degree distribution of **events** in the reduced event graph.



PRR Supplement, Eq. (4).

Generating functions and arrival bias

$$G(x,y) \quad = \quad \sum_{i,o=0}^{2} P(i,o)x^i y^o$$

Generating function of the joint in/out-degree distribution.

$$z = \left.\frac{\partial G}{\partial x}\right|_{x=y=1} = \left.\frac{\partial G}{\partial y}\right|_{x=y=1} = \langle k_{\text{in}} \rangle = \langle k_{\text{out}} \rangle.$$

$$G_0^{\text{out}}(y) \;=\; G(1,y)$$

Out-degree distribution of a **uniformly chosen** event.

$$G_1^{\text{out}}(y) \;=\; \frac{1}{z}\left.\frac{\partial G(x,y)}{\partial x}\right|_{x=1}$$

Out-degree distribution seen after **arriving** at an event along a directed edge.

The derivative is with respect to x, the in-degree variable, because events with more incoming edges are reached and sampled more often.

PRR Supplement, Eqs. (5)–(7).

Three different degree distributions

Object	Distribution	Interpretation
Static network	p_k	degree of a uniformly chosen physical node
Static network, after following an edge	$q_k = \frac{(k+1)p_{k+1}}{\langle k \rangle}$	if the reached physical node has degree $k+1$, one edge was used for arrival and k static edges remain
Directed event graph, after following an incoming edge	$Q_{\text{out}}(o)$	number o of future event continuations; no outgoing edge is subtracted

The threshold $\widehat{q}_{\text{out}} = 1$ refers to the **arrival-biased event-graph distribution**, not to the degree of the static base network.

Static excess degree: formula and examples

A uniformly chosen node has degree k with probability p_k. The endpoint of a uniformly chosen edge is instead sampled with the size-biased probability

$$\Pr(K = k \mid \text{arrive along an edge}) = \frac{k p_k}{\langle k \rangle}.$$

After using the arrival edge, the remaining degree is $j = k - 1$. Therefore

$$q_j = \frac{(j+1)p_{j+1}}{\langle k \rangle}, \qquad \langle q \rangle = \frac{\langle k^2 \rangle - \langle k \rangle}{\langle k \rangle}.$$

9-regular base network

$p_9 = 1$, so every arrival leaves

$$\langle q \rangle = 9 - 1 = 8.$$

ER base network, $\langle k \rangle = 8$

For a Poisson degree distribution, $\langle k^2 \rangle = \langle k \rangle^2 + \langle k \rangle$, hence

$$\langle q \rangle = \langle k \rangle = 8.$$

Both examples concern the static base network. They do not state the event-graph branching threshold.

Directed excess out-degree

$Q_{\mathrm{out}}(o)$ is the probability of finding o outgoing continuations after arriving at an event along a directed edge.

$$Q_{\mathrm{out}}(o) = [y^o] G_1^{\mathrm{out}}(y) = \frac{1}{o!} \frac{d^o G_1^{\mathrm{out}}(y)}{dy^o}\bigg|_{y=0}$$

Coefficient of y^o: probability of o outgoing continuations after arrival.

Equivalently, using the joint event-degree distribution,

$$\boxed{Q_{\mathrm{out}}(o) = \frac{\sum_i iP(i,o)}{\langle k_{\mathrm{in}} \rangle}}.$$

The factor i is the arrival bias: events with larger in-degree have more incoming edges and are therefore sampled more often. Thus $Q_{\mathrm{out}}(0)$, $Q_{\mathrm{out}}(1)$ and $Q_{\mathrm{out}}(2)$ are the branching probabilities used by mean field.

PRR Supplement, Eqs. (5)–(8).

Mean excess out-degree and threshold

The mean number of outgoing continuations after arrival is

$$\boxed{\widehat{q}_{\text{out}} = \sum_{o} o Q_{\text{out}}(o) = \frac{\langle k_{\text{in}} k_{\text{out}} \rangle}{\langle k_{\text{in}} \rangle}}.$$

Equivalently,

$$\widehat{q}_{\text{out}} = \langle k_{\text{out}} \rangle + \frac{\text{Cov}(k_{\text{in}}, k_{\text{out}})}{\langle k_{\text{in}} \rangle}.$$

Static-base-network facts

A 9-regular network has static excess degree 8. An ER network with $\langle k \rangle = 8$ also has mean static excess degree 8.

Event-graph criterion

In $\widehat{D}_{\delta t}$, $o \in \{0, 1, 2\}$ and the locally tree-like branching threshold is $\widehat{q}_{\text{out}} = 1$.

The number 8 describes candidate static links; the threshold 1 describes arrival-biased event-graph branching.

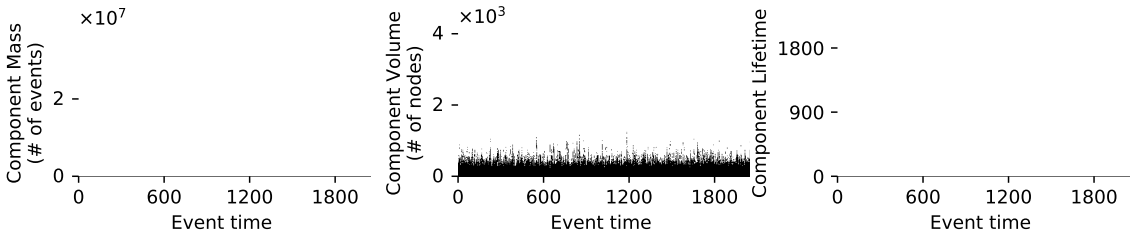
PRR Supplement, Eqs. (5)–(8).

Analytical roadmap: from temporal rules to mean field

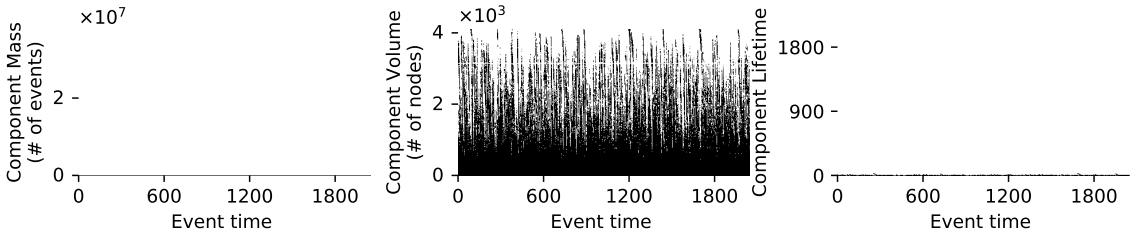
1. **Choose an event:** $e = (l, r, t_0)$.
2. **Static topology:** sample q_l, q_r: candidate adjacent physical links.
3. **Temporal dynamics:** original link uses $\mathcal{T}$; adjacent links use $\mathcal{R}$.
4. **Conditional branching:** compute P_0, P_2 and $P_1 = 1 - P_0 - P_2$ for fixed q_l, q_r.
5. **Average over P_q:** obtain $P(k_{\text{in}}, k_{\text{out}})$.
6. **Generating function:** $G(x, y) = \sum P(i, o) x^i y^o$.
7. **Arrival bias:** $G_1^{\text{out}}(y)$ is the out-degree law after following a directed edge.
8. **Branching statistics:** extract $Q_{\text{out}}(0, 1, 2)$ and $\widehat{q}_{\text{out}} = \sum_o o Q_{\text{out}}(o)$.
9. **Mean-field closure:** replace the event graph by a random locally tree-like digraph with the same $P(i, o)$.
10. **Rate equation:** $\partial_t \rho = (Q_2 - Q_0)\rho - (Q_1 + 2Q_2)\rho^2$. Then $\tau = \widehat{q}_{\text{out}} - 1$ and $\widehat{q}^c_{\text{out}} = 1$.

Synthetic transition at event level

Below
$\delta t = 0.080$
finite clusters



Critical
$\delta t \simeq 0.088$
broad range



From left to right: event-wise mass M_e, volume V_e and lifetime T_e. Near criticality, the distribution spans many scales.

PRR Supplemental Material, Fig. 2.

Directed-percolation exponents used in the papers

Exponent	Scaling law	Meaning
β	$\rho_{\mathrm{stat}} \sim \tau^{\beta}$	stationary density
β'	$P_{\infty} \sim \tau^{\beta'}$	ultimate survival
α	$\rho(t) \sim t^{-\alpha}$	critical density decay
δ	$P(t) \sim t^{-\delta}$	critical survival decay
γ	$M \sim (-\tau)^{-\gamma}$	mean cluster mass
$\nu_{\parallel}$	$\xi_{\parallel} \sim \lvert\tau\rvert^{-\nu_{\parallel}}$	temporal correlations
$\nu_{\perp}$	$\xi_{\perp} \sim \lvert\tau\rvert^{-\nu_{\perp}}$	spatial correlations

Rapidity reversal gives $\beta = \beta'$ and therefore $\alpha = \delta = \beta/\nu_{\parallel}$ for DP.

Scaling relations and mean-field values

Dynamic scaling

$$\alpha = \frac{\beta}{\nu_\parallel}, \qquad \delta = \frac{\beta'}{\nu_\parallel}, \qquad z = \frac{\nu_\parallel}{\nu_\perp}.$$

Cluster-mass hyperscaling, $d < d_c$

$$\gamma = \nu_\parallel + d\nu_\perp - \beta - \beta'.$$

Rapidity reversal in DP

$$\beta = \beta' \implies \alpha = \delta.$$

Upper critical dimension

$$d_c = 4.$$

Mean field: $\beta = \beta' = 1$, $\nu_\parallel = 1$, $\nu_\perp = 1/2$, $\alpha = \delta = 1$, $\gamma = 1$.

At $d = d_c$ logarithmic corrections can occur; above d_c, naive hyperscaling must be modified.

Explicit solution of the mean-field equation

For $\partial_t \rho = \tau\rho - g\rho^2$ and $\rho(0) = \rho_0$,

$$\rho(t) = \frac{\tau\rho_0 e^{\tau t}}{\tau + g\rho_0(e^{\tau t} - 1)} \qquad (\tau \neq 0).$$

$$\begin{aligned} \tau < 0: &\quad \rho(t) \longrightarrow 0, \\ \tau > 0: &\quad \rho(t) \longrightarrow \tau/g, \\ \tau = 0: &\quad \rho(t) = \frac{\rho_0}{1 + g\rho_0 t} \sim t^{-1}. \end{aligned}$$

Here $g = \widehat{q}_{\mathrm{out}}$ for the reduced event graph.

Why the critical excess degree is one

In the locally tree-like mean-field picture, following an edge reaches an event with an average of $\widehat{q}_{\text{out}}$ outgoing continuations.

$\widehat{q}_{\text{out}} < 1$	$\widehat{q}_{\text{out}} = 1$	$\widehat{q}_{\text{out}} > 1$
branching dies out	critical branching	a giant out-component can emerge

Therefore, the transition occurs at

$$\widehat{q}^{c}_{\text{out}} = 1, \qquad \tau = \widehat{q}_{\text{out}} - 1.$$

Rate equation: derivation line by line

Write $Q_j \equiv Q_{\rm out}(j)$, with $Q_0 + Q_1 + Q_2 = 1$. A reached event changes the number of active branches by

$$(-1)Q_0 + 0Q_1 + (+1)Q_2 = Q_2 - Q_0.$$

The expected number of attempted continuations is

$$Q_1 + 2Q_2 = \widehat{q}_{\rm out}.$$

If a fraction ρ is already occupied, mergers generate the quadratic correction. Therefore

$$\begin{aligned} \partial_t \rho &= (Q_2 - Q_0)\rho - (Q_1 + 2Q_2)\rho^2, \\ Q_2 - Q_0 &= Q_2 - (1 - Q_1 - Q_2) \\ &= Q_1 + 2Q_2 - 1 = \widehat{q}_{\rm out} - 1 \equiv \tau, \end{aligned}$$

and hence

$$\boxed{\partial_t \rho = \tau\rho - \widehat{q}_{\rm out}\rho^2.}$$

PRR, Eq. (1), and Supplemental Material, Eq. (8).

Generating functions for the mean cluster mass

$$H_1(z) = zG_1^{\text{out}}(H_1(z)),$$
$$H_0(z) = zG_0(1, H_1(z)).$$

Differentiating at $z = 1$ in the subcritical phase:

$$H_1'(1) = \frac{1}{1 - G_1^{\text{out}\prime}(1)} = \frac{1}{1 - \widehat{q}_{\text{out}}},$$

and

$$M = H_0'(1) = 1 + \langle k_{\text{out}} \rangle H_1'(1) \propto \frac{1}{1 - \widehat{q}_{\text{out}}} = \frac{1}{-\tau}.$$

Hence $\gamma = 1$.

Rapidity reversal: density and survival

Forward spreading

$P(t)$ follows the time span of out-components

Backward ancestry

$\rho(t)$ is determined by in-components

If the event-graph ensemble is statistically invariant under time reversal,

$$p_{i,o}^{\text{in,out}} = p_{o,i}^{\text{in,out}},$$

$$\rho(t) = P(t), \qquad \rho_{\text{stat}}(\tau) = P_\infty(\tau), \qquad \beta = \beta'.$$

Why expect the directed-percolation universality class?



The Janssen–Grassberger criterion suggests DP for a continuous transition with:

Ingredients present here

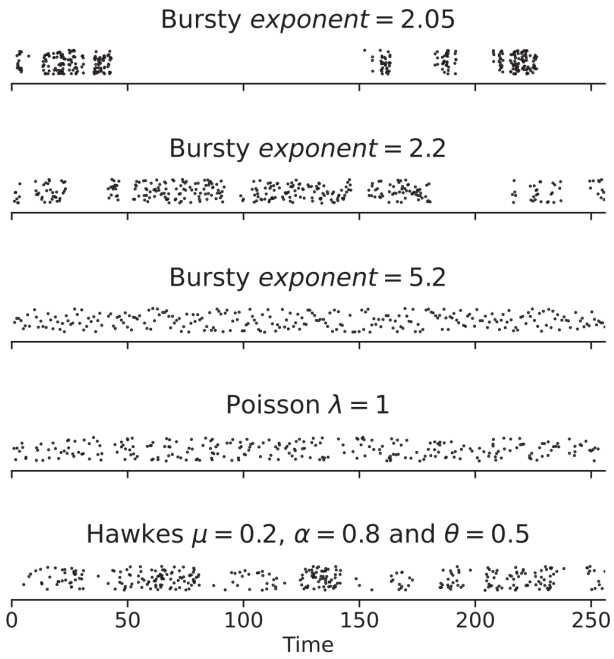
- a preferred direction: increasing time;
- an absorbing state: no reachable activity;
- a scalar activity density ρ;
- local propagation through shared nodes.

Possible reasons to leave DP

- extra symmetries or conserved fields;
- quenched disorder or long-range coupling;
- strong memory and long temporal correlations;
- modularity and multiple competing scales.

This is a universality **expectation**, not a proof. The PRE finite-size collapses test whether it survives structural and temporal heterogeneity.

Temporal activation processes



Poisson
Memoryless process with exponentially distributed inter-event times.

Bursty renewal
Independent inter-event times drawn from a broad, power-law distribution produce bursts and long gaps.

Hawkes
Self-exciting: an event temporarily increases the future rate, creating temporal correlations.

All processes are normalized to $\langle \Delta t \rangle = 1$: differences therefore reflect **temporal organization**, not the mean activity rate.

PRE, Fig. 2.

Poisson

$$p_{\mathrm{P}}(\Delta t) = \lambda e^{-\lambda \Delta t}, \qquad \Delta t \geq 0, \qquad \lambda = 1.$$

Bursty renewal

$$p_{\mathrm{B}}(\Delta t) = (\gamma - 1)\Delta t_{\mathrm{min}}^{\gamma-1} \Delta t^{-\gamma}, \qquad \Delta t \geq \Delta t_{\mathrm{min}}, \qquad \Delta t_{\mathrm{min}} = \frac{\gamma - 2}{\gamma - 1}.$$

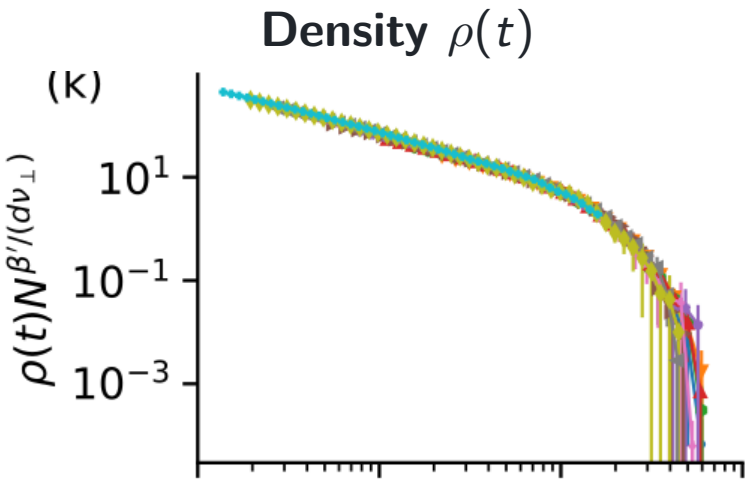
Hawkes

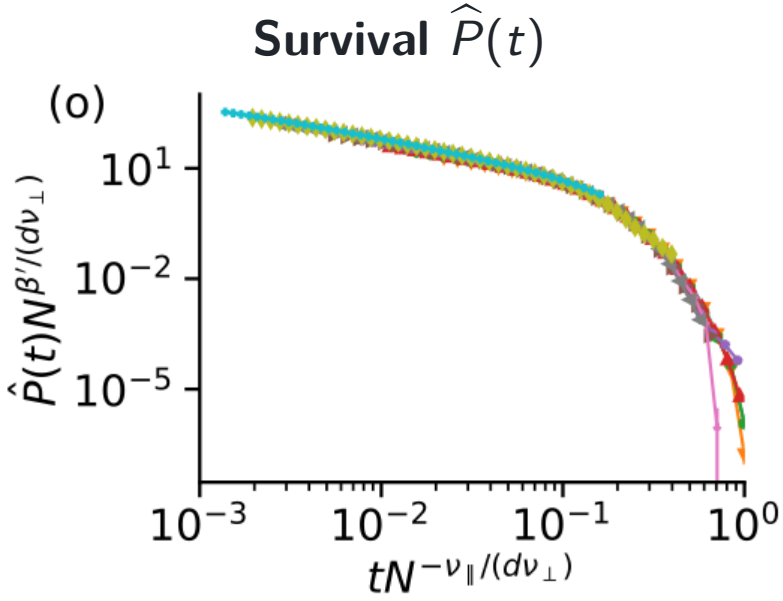
$$\lambda^*(t) = \mu + \alpha\theta \sum_{t_i < t} e^{-\theta(t - t_i)}, \qquad \overline{\lambda} = \frac{\mu}{1 - \alpha} = 1.$$

$\langle \Delta t \rangle$ = 1 in all three cases.

PRE, Sec. III A and Eq. (8).

What an actual finite-size collapse looks like





Example: random 9-regular network with bursty renewal dynamics. Curves for $N = 2^8, \ldots, 2^{17}$ collapse after using the critical point and the mean-field DP rescaling.

PRE, Fig. 5(k) and 5(o).

Why τ is approximately linear in δt near criticality

$$\tau(\delta t) = \widehat{q}_{\mathrm{out}}(\delta t) - \widehat{q}^{c}_{\mathrm{out}}, \qquad \text{mean field: } \widehat{q}^{c}_{\mathrm{out}} = 1.$$
$$\text{At } \delta t = \delta t_c,\ \widehat{q}_{\mathrm{out}}(\delta t_c) = \widehat{q}^{c}_{\mathrm{out}}, \text{ hence } \tau = 0.$$

Expand the smooth function $\widehat{q}_{\mathrm{out}}(\delta t)$ to first order around δt_c:

$$\widehat{q}_{\mathrm{out}}(\delta t) \simeq \widehat{q}_{\mathrm{out}}(\delta t_c) + \left.\frac{d\widehat{q}_{\mathrm{out}}}{d\delta t}\right|_{\delta t_c} (\delta t - \delta t_c).$$

$$\tau \simeq A(\delta t - \delta t_c), \qquad A = \left.\frac{d\widehat{q}_{\mathrm{out}}}{d\delta t}\right|_{\delta t_c} > 0.$$

Local Taylor expansion only—not a universality assumption.

larger waiting-time cutoff δt → more allowed temporal continuations → larger effective branching $\widehat{q}_{\mathrm{out}}$

PRE, Appendix and Fig. 7.

Why finite-size scaling is needed

A finite active system can eventually reach the absorbing state through a fluctuation.

The corresponding crossover time grows with system size:

$$t_f \sim \ell^z, \qquad z = \frac{\nu_\parallel}{\nu_\perp}, \qquad N \sim \ell^d.$$

At the correct δt_c and with the correct exponents, rescaled curves for different N collapse onto common scaling functions.

The collapse simultaneously tests the estimated threshold and the proposed universality class.

Where the finite-size scaling form comes from

1. Correlation scales

$$\xi_\perp \sim |\tau|^{-\nu_\perp}, \qquad \xi_\parallel \sim |\tau|^{-\nu_\parallel}.$$

Eliminating $|\tau|$ gives

$$\xi_\parallel \sim \xi_\perp^{\nu_\parallel/\nu_\perp}.$$

Hence a spatial scale ℓ has the characteristic time

$$t \sim \ell^{\nu_\parallel/\nu_\perp}.$$

At criticality in a finite system, $\xi_\perp$ cannot exceed ℓ.

2. Density at finite size

Since

$$\rho \sim \xi_\perp^{-\beta/\nu_\perp},$$

replace $\xi_\perp$ by the limiting scale ℓ and retain time through the scale-invariant ratio:

$$\rho(t,\ell) = \ell^{-\beta/\nu_\perp} F\left(t\ell^{-\nu_\parallel/\nu_\perp}\right).$$

Therefore

$$\ell^{\beta/\nu_\perp}\rho = F\left(t\ell^{-\nu_\parallel/\nu_\perp}\right).$$

Using $N \sim \ell^d$ gives

$$N^{\beta/(d\nu_\perp)}\rho = F\left(tN^{-\nu_\parallel/(d\nu_\perp)}\right).$$

If the threshold and exponents are correct, this rescaling absorbs finite-size effects and trajectories for different N collapse onto one scaling function.

PRE, Eqs. (4)–(6).

Scale invariance: the PRE equations

Multiplicative rescaling around the critical point

$$t \mapsto \lambda^{-\nu_\parallel} t, \qquad \ell \mapsto \lambda^{-\nu_\perp} \ell,$$

$$\rho \mapsto \lambda^{\beta} \rho, \qquad P \mapsto \lambda^{\beta'} P, \qquad h \mapsto \lambda^{\sigma} h, \qquad \chi \mapsto \lambda^{-\gamma} \chi.$$

Apply scale invariance to the density

$$\rho = f(t, \ell) \quad \Longrightarrow \quad \lambda^{\beta} \rho = f\left(\lambda^{-\nu_\parallel} t, \lambda^{-\nu_\perp} \ell\right).$$

Choose $\lambda = \ell^{1/\nu_\perp}$

$$\boxed{\ell^{\beta/\nu_\perp} \rho = f\left(\ell^{-\nu_\parallel/\nu_\perp} t, 1\right) = F\left(\ell^{-\nu_\parallel/\nu_\perp} t\right)}.$$

PRE, Eqs. (4)–(6).

Collapse error in Fig. 3

1. Candidate δt
choose one possible waiting-time cutoff

2. Simulate several N
obtain one trajectory for each system size

3. Rescale trajectories
apply the finite-size-scaling axes

4. Common range
compare curves only where all sizes overlap

Over the common range, calculate a logarithmic RMS error separately for each $X \in \{\widehat{P}, \rho, M, V\}$:

$$E_X(\delta t) = \left[\frac{1}{n_{\mathrm{pts}}} \sum_{N,\, x \in \mathcal{I}} \left(\log X_N^{\mathrm{sc}}(x) - \log \overline{X}^{\mathrm{sc}}(x) \right)^2 \right]^{1/2}$$

Schematic form of the log-RMS metric described in PRE.

where $\mathcal{I}$ is the common rescaled-time interval and $\overline{X}^{\mathrm{sc}}$ is the average trajectory on that interval.

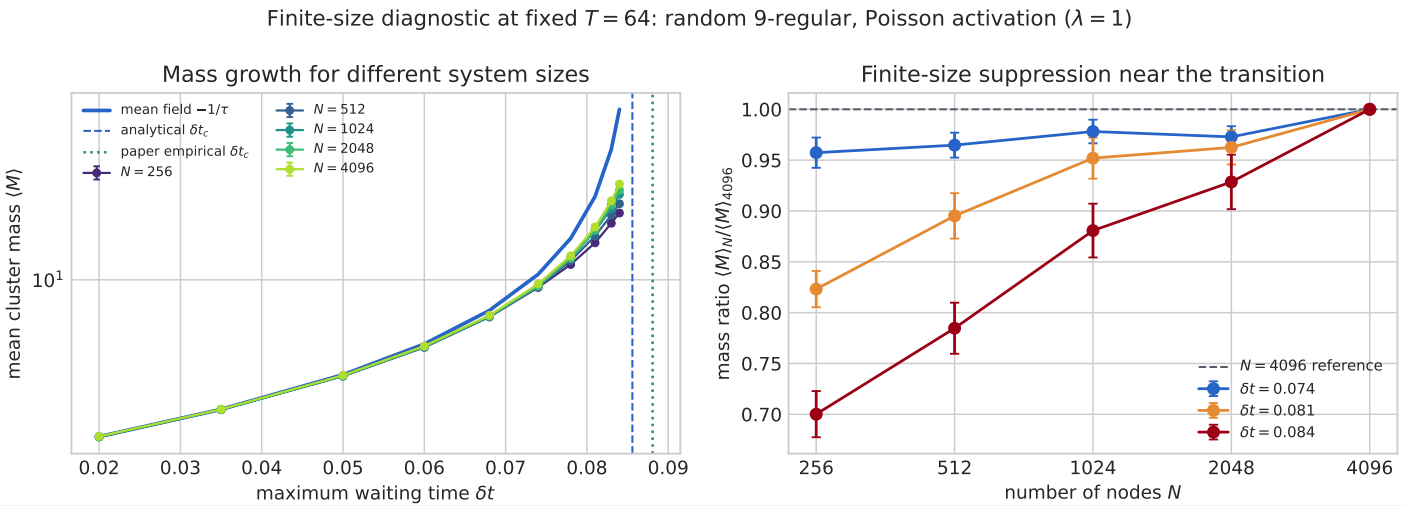
$E_{\mathrm{tot}} = E_{\widehat{P}} + E_\rho + E_M + E_V$
each Fig. 3 point
already compares several N

→

$\delta t_c = \arg\min_{\delta t} E_{\mathrm{tot}}(\delta t)$
the sharp minimum in Fig. 3

PRE, Fig. 3 and Sec. III B.

Finite-size diagnostic of our simulation



At $\delta t = 0.084$, $M_{256}/M_{4096} = 0.70$; at $\delta t = 0.074$, the ratio is 0.96.

Since $T = 64$ is fixed, this is a diagnostic comparison—not a full finite-size-scaling collapse.

Our computation: exact workflow



1. Generate a random 9-regular base network with $N = 4096$ nodes.
2. On every static edge, generate an independent stationary Poisson activation process with $\langle \Delta t \rangle = 1$ over $T = 64$.
3. Connect each event to the first later event at each endpoint. This gives a reachability-preserving structure with $k_{\mathrm{out}} \leq 2$.
4. For each chosen δt, retain only transitions with waiting time $\Delta t \leq \delta t$.
5. Sample 1024 source events away from the temporal boundaries, compute each exact reachable out-component by graph search, and average its mass.
6. Average the result over 128 independent temporal-network realizations.

Fit $M \propto (\delta t_c - \delta t)^{-\gamma_{\mathrm{eff}}}$ only in the subcritical window $0.050 \leq \delta t \leq 0.081$, using the PRE value $\delta t_c = 0.08808$.

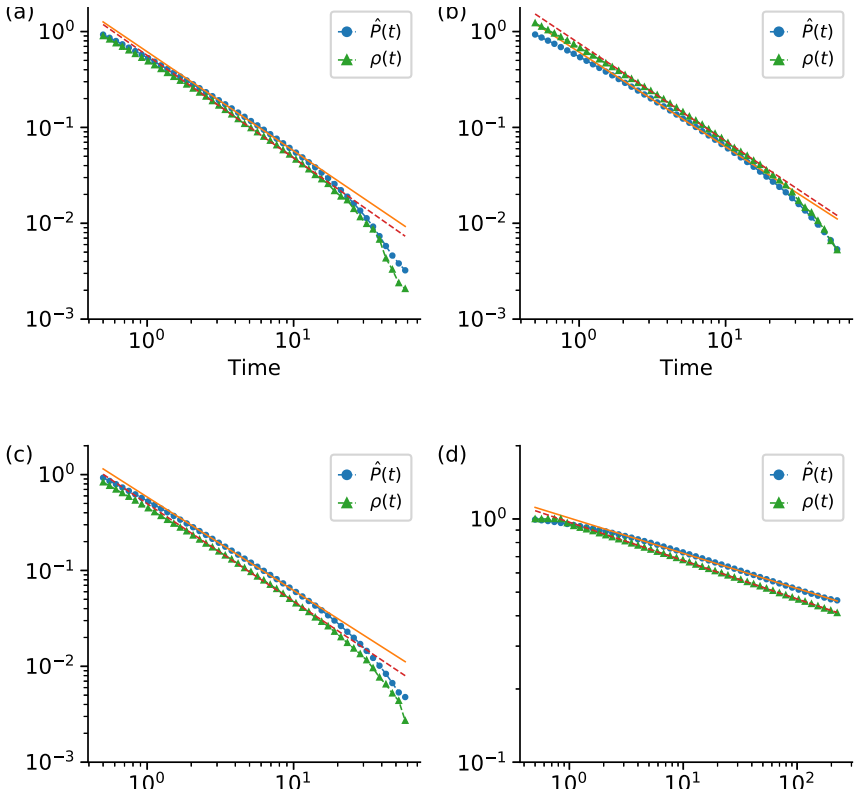
Our computation vs. the PRE

	Our illustration	**PRE study**
Model	9-regular, Poisson	26 structural/temporal combinations
Sizes	main $N = 4096$; diagnostic 256–4096	10 sizes, $N = 2^8$–2^{17}
Statistics	128 realizations, 1024 sampled sources	256–4096 realizations; all sources processed in one pass
Largest realization	about 1.18×10^6 events	about 3.7×10^7 events
Observable	mean mass M	M, V, P, ρ and other single-/many-seed observables
Critical point	imported from PRE	estimated by joint finite-size-collapse error
Conclusion	consistency check: $\gamma_{\mathrm{eff}} \simeq 1.09$	threshold and universality-class test

More realizations reduce sampling uncertainty as $R^{-1/2}$; larger N postpones the finite-size cutoff. Neither alone replaces a multi-size collapse.

PRE, Secs. II B and III A; our simulation metadata.

Direct critical-time power-law fits



At the estimated δt_c, the slopes of $\rho(t) \sim t^{-\alpha}$ and $\widehat{P}(t) \sim t^{-\delta}$ provide a direct exponent check.

PRE, Fig. 6.

Representative numerical exponents

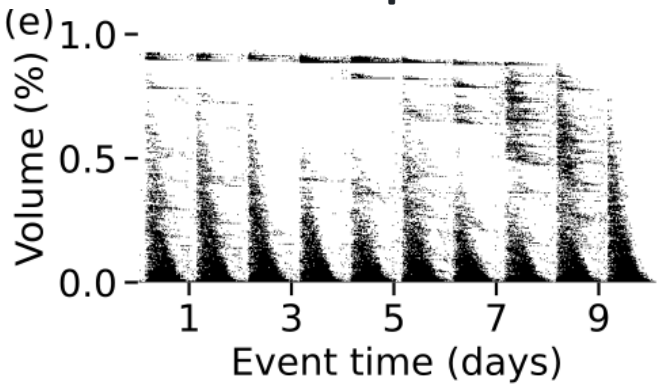
Configuration	δt_c	$\widehat{\alpha}$	$\widehat{\delta}$
Erdős–Rényi, Poisson	0.08421	1.0702	1.0338
random 9-regular, Poisson	0.08808	1.0096	0.9947
1D lattice, Poisson	0.9919	0.1583	0.1456
2D lattice, Poisson	0.28428	0.4109	0.3922
3D lattice, Poisson	0.15375	0.7229	0.6899
4D lattice, Poisson	0.1045	1.0077	0.9870

Expected DP values: 0.15946 (1D), 0.450 (2D), 0.732 (3D), and leading mean-field value 1 for $d \geq 4$; logarithmic corrections may occur at $d = 4$.

PRE, Table I.

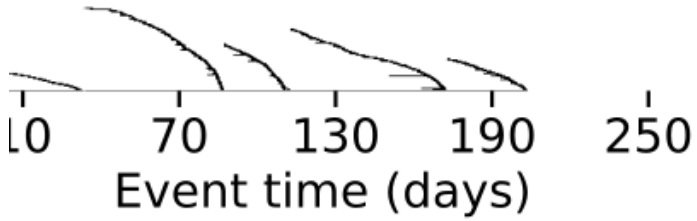
Event-level profiles reveal organization

Air transport



Repeated daily bands reflect the timetable and overnight interruptions.

Twitter mentions



Relative reachable volume: separated components coexist with occasional much larger sets.

Averages reveal the transition; event-level profiles reveal temporal organization. Helsinki and mobile calls show complementary wing-like finite-window envelopes.

PRR, Fig. 3(e)–(h).

Real-network transition scales



Network	Observed onset	Analytical estimate
US air transport	~ 470 s	500 s
Helsinki public transport	~ 670 s	488 s
Twitter mentions	~ 25 min	119.1 h
mobile-phone calls	~ 7.5 h	22.5 h

The discrepancy for some real networks is consistent with structural correlations absent from the random-network approximation. Multiple susceptibility peaks reveal additional characteristic timescales.

PRR, Fig. 3 and caption.

Main references


A. Badie-Modiri, A. K. Rizi, M. Karsai, and M. Kivelä,
"Directed percolation in temporal networks,"
Phys. Rev. Research **4**, L022047 (2022).


A. Badie-Modiri, A. K. Rizi, M. Karsai, and M. Kivelä,
"Directed percolation in random temporal network models with heterogeneities,"
Phys. Rev. E **105**, 054313 (2022).


H. Hinrichsen,
"Non-equilibrium critical phenomena and phase transitions into absorbing states,"
Adv. Phys. **49**, 815–958 (2000).